Inorganic CUME
April-9-2021

(Note: Answer 5 of the 10 questions below. Only first 5 will be graded)

1. (***20 points***) For the $M(C_5H_5)_2$, where the M = Fe, Co, and Ni:
   i) Show d-orbital electrons of the M in the LUMO and HOMO to predict which compound(s) may have the speak-broadening in the NMR spectrum.
   ii) Discuss the difference in a property (or properties) for the three compounds.

2. (***20 points***) More for the sandwich compounds:
   i) Draw the $Ni(COT)_2$ structure, and a multideck sandwich structure that looks like a metal organic framework (or MOF), respectively.
   ii) Explain whether the 18-elctron rule can work for the two special sandwich compounds.

3. (***20 points***) For the half-sandwich compounds:
   i) Draw a stable half-sandwich structure of Cp* with V, and that with Co, respectively.
   ii) Draw a half sandwich structure in which the Cp* is replaced by a cyclopentadiene. Compare the reactivity of the Cp*-containing structure with that of the cyclopentadiene-containing structure.

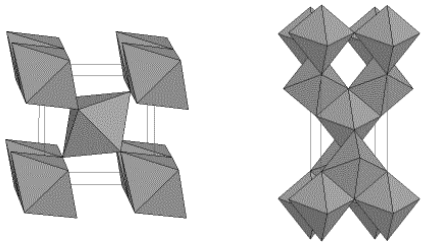
4. (***20 points***) Based on the Latimar Diagram below,

E^o(V)	*0.564*		*1.44*		*-0.70*		*1.51*		*-1.18*	
MnO_4^-	→	MnO_4^{2-}	→	MnO_2	→	Mn^{3+}	→	Mn^{2+}	→	Mn

   i. Discuss the coordination bonding of a ligand of your choice with Mn^{2+} and explain why the bonding may not be stable if using Mn^{3+}.
   ii. If slowly mixing a small amount of a reducing organic agent into a dilute aqueous solution of the purple $KMnO_4$, a green liquid can form on top of the solution and then quickly disappear upon stirring, which can result in a brown precipitate as a product. Explain the chemical reaction(s) in this entire process.

5. (***20 points***) Pick any five in the list below to discuss briefly how you could use each to characterize what for an organometallic compound (or a coordination compound) mentioned in the above.
   a) FTIR
   b) NMR
   c) GC-MS
   d) ICP-MS
   e) XRD
   f) SEM
   g) Raman
   h) AA
   i) TEM
   j) EPR
   k) UV-vis
   l) XPS
   m) CV
   n) AFM
   o) Fluorescent microscope
   p) Laser scattering

6. (***20 points***) TiO_2 has two structures (rutile and anatase) below.



Rutile Anatase

i) Which is more thermally stable and electrically conductive than the other? Explain why so.
ii) Which one can be a photocatalyst and which as a sunscreen skin lotion component? Explain why.

7. (***20 points***) For each material below, draw the unit cell, and show the (122) lattice plane in the unit cell:
   i) gold crystal
   ii) NaCl crystal
   iii) CsCl crystal
   iv) ZnO hexagonal crystal

8. (***20 points***) The average pore size of the surgical mask and N95 mask is around 10–20 microns, which may not be able to block 100% of SARS-CoV-2 virus particle that has an average diameter of 0.12 micron, as the pore size smaller than 0.12 micron may not allow enough oxygen to let you breathe properly. Find a **practical method** for doing some chemistry tricks effectively on these facemasks, to let you wear your new facemask that can quickly deactivate the coronavirus once the virus particle sticks on the facemask surface.

9. (***20 points***) Some new COVID vaccine formulations use nanomaterials. If you were asked to develop a new vaccine for oral or intravenous injection use, choose your best choice of the nanomaterials and explain why this choice to the best of your knowledge can potentially meet the FDA criteria.

10. (***20 points***) In the materials chemistry for energy storage, a long-unmet challenge is to find a robust and safe absorbent material, easy for mass-production at low-cost, that can store natural gas or hydrogen gas at ambient temperature and pressure. Propose a new material and explain why the structure and surface chemistry can help meet the need to effectively store and release which gas, potentially.